

Introduction to Relations and Functions

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Relations

A **relation** is a set of ordered pairs (x, y) , where x is the **input** and y is the **output**.

Functions

A **function** is a relation in which each input is paired with **exactly one** output.

Not a Function

A relation is **not a function** if at least one input has two or more different outputs.

Representing Relations

Relations can be represented as:

Ordered pairs

Tables of values

Graphs

Mapping diagrams

Checking a Relation

To decide whether a relation is a function, check whether any input value repeats with different output values.

Examples

Example 1

Determine whether the relation is a **function** or **mere relation**.

$$\{(1, 2), (2, 4), (3, 6), (4, 8)\}$$

Example 2

Determine whether the relation is a **function** or **mere relation**.

$$\{(2, 5), (2, 7), (4, 9)\}$$

Example 3

Determine whether the relation is a **function** or **mere relation**.

$$\{(-1, 3), (0, 1), (1, 3)\}$$

Example 4

Determine whether the relation is a **function** or **mere relation**.

$$\{(0, 0), (1, 1), (1, 2), (2, 4)\}$$

Practice Exercises

Practice 1

Determine whether the relation is a **function** or **mere relation**.

$$\{(1, 2), (2, 4), (3, 6), (4, 8)\}$$

Practice 2

Determine whether the relation is a **function** or **mere relation**.

$$\{(2, 5), (2, 7), (4, 9)\}$$

Practice 3

Determine whether the relation is a **function** or **mere relation**.

$$\{(-1, 3), (0, 1), (1, 3)\}$$

Practice 4

Determine whether the relation is a **function** or **mere relation**.

$$\{(0, 0), (1, 1), (1, 2), (2, 4)\}$$

Practice 5

Determine whether the relation is a **function** or **mere relation**.

Table: $x = \{1, 2, 3, 4\}$, $y = \{5, 6, 7, 8\}$.

Practice 6

Determine whether the relation is a **function** or **mere relation**.

Table: $x = \{1, 1, 2, 3\}$, $y = \{4, 5, 6, 7\}$.

Practice 7

Determine whether the relation is a **function** or **mere relation**.

$$y = 2x + 1$$

for $x = \{-1, 0, 1, 2\}$.

Practice 8

Determine whether the relation is a **function** or **mere relation**.

$$y = \frac{1}{x}$$

for $x = \{1, 2, -1, 1\}$.

Practice 9

Determine whether the relation is a **function** or **mere relation**.

Mapping:

$$1 \rightarrow 3, \quad 2 \rightarrow 5, \quad 3 \rightarrow 7$$

Practice 10

Determine whether the relation is a **function** or **mere relation**.

Mapping:

$$1 \rightarrow 4, \quad 1 \rightarrow 6, \quad 2 \rightarrow 8$$

Activities

Activity 1

Determine whether the relation is a **function** or **mere relation**.

$$\{(1, 3), (2, 6), (3, 9), (4, 12)\}$$

Activity 2

Determine whether the relation is a **function** or **mere relation**.

$$\{(3, 4), (3, 6), (5, 8)\}$$

Activity 3

Determine whether the relation is a **function** or **mere relation**.

$$\{(-2, 1), (0, 5), (2, 1)\}$$

Activity 4

Determine whether the relation is a **function** or **mere relation**.

$$\{(1, 2), (2, 4), (2, 6), (3, 8)\}$$

Activity 5

Determine whether the relation is a **function** or **mere relation**.

Table: $x = \{2, 4, 6, 8\}$, $y = \{1, 3, 5, 7\}$.

Activity 6

Determine whether the relation is a **function** or **mere relation**.

Table: $x = \{0, 0, 1, 2\}$, $y = \{2, 3, 4, 5\}$.

Activity 7

Determine whether the relation is a **function** or **mere relation**.

$$y = 3x - 2$$

for $x = \{-2, -1, 0, 1\}$.

Activity 8

Determine whether the relation is a **function** or **mere relation**.

$$y = \frac{2}{x}$$

for $x = \{1, 2, 2, -1\}$.

Activity 9

Determine whether the relation is a **function** or **mere relation**.

Mapping:

$$2 \rightarrow 4, \quad 4 \rightarrow 6, \quad 6 \rightarrow 8$$

Activity 10

Determine whether the relation is a **function** or **mere relation**.

Mapping:

$$3 \rightarrow 5, \quad 3 \rightarrow 7, \quad 4 \rightarrow 9$$

Thank you!